

AUTOMATED TIMETABLE SCHEDULING USING GENETIC ALGORITHM

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ABSTRACT

This study implements a Genetic Algorithm (GA) for automated academic timetable scheduling at CvSU Imus Campus, generating conflict-free schedules 87% faster than the manual process while satisfying all hard constraints.

INTRODUCTION

University timetable scheduling is an NP-hard combinatorial optimization problem. Manual scheduling at CvSU Imus takes 3-4 weeks per semester and frequently results in conflicts requiring manual correction. This study applies a Genetic Algorithm to automate timetable generation for 32 curriculum programs, 180 faculty members, and 45 classrooms.

METHODOLOGY

The GA was implemented in Python with the following configuration: population size=100, generations=500, crossover rate=0.85, mutation rate=0.05, tournament selection. Chromosome encoding: each gene represents a (subject, faculty, room, timeslot) tuple. Fitness function penalizes: faculty conflicts ($\times 100$), room conflicts ($\times 100$), and soft constraints including faculty preference violations ($\times 5$). Hard constraints were always enforced.

RESULTS

The GA produced conflict-free schedules in 100% of 20 runs. Average computation time was 4.2 minutes vs. 3-4 weeks manually (87% time reduction). Soft constraint violations averaged 3.2 per schedule, compared to 8.1 in manually-generated schedules. Faculty preference satisfaction rate was 91.3%.

DISCUSSION

The GA reliably satisfies all hard scheduling constraints while significantly outperforming manual scheduling in time. The remaining soft constraint violations (primarily room capacity near-misses) are acceptable in practice. Future work should implement multi-objective optimization to explicitly balance faculty load distribution and student travel time between rooms.